

Analysis of Bike Sharing Data

Introduction

Bike-sharing systems have become an important mode of urban transportation, providing an environmentally friendly and convenient alternative to private vehicle use.

Understanding the factors that influence bike rental demand is valuable for system planning, resource allocation, and service improvement.

This report analyses a bike-sharing dataset containing daily rental information collected during 2011 and 2012. The objective is to examine the distribution of bike rentals and investigate whether rental demand differs according to year, season, weather conditions, holiday status, and working-day status.

The analysis was conducted in R and combines descriptive statistics with hypothesis testing to identify significant patterns within the data.

Dataset Description

The dataset contains 515 observations and 12 variables. The response variable is the total number of daily bike rentals (cnt), while the remaining variables describe temporal and environmental conditions associated with each observation.

The variables included in the analysis are:

Variable	Description
season	Winter, Spring, Summer, Fall
yr	Year (2011 or 2012)
mnth	Month
holiday	Indicates whether the day was a holiday
weekday	Day of the week
workingday	Indicates whether the day was a working day
weathersit	Weather condition
cnt	Total bike rentals
temp1	Temperature (°C)
atemp1	Feeling temperature (°C)
hum1	Humidity (%)
windspeed1	Wind speed

The primary variable of interest is the total number of bike rentals (cnt), which represents overall demand for the bike-sharing service.

Data Preparation

The data were imported from an SPSS (.sav) file into R. Variables stored as labelled values were converted into factors so that categorical analyses could be performed correctly.

A missing value assessment was conducted before analysis. No missing observations were identified in any of the variables.

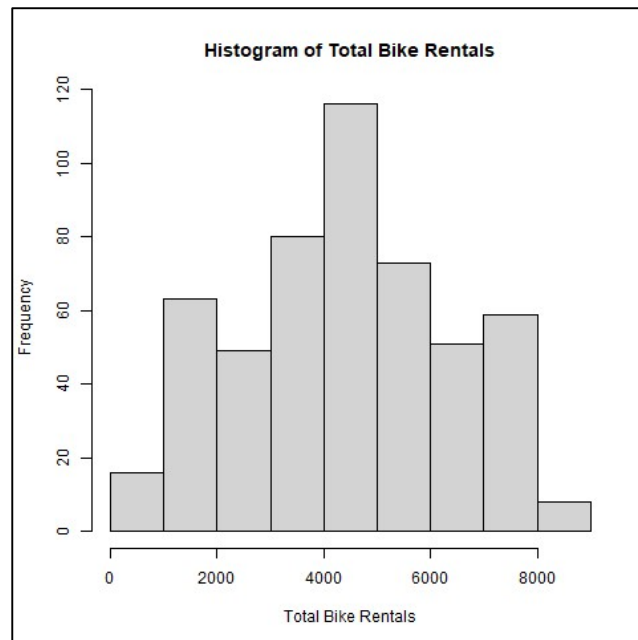
Following data preparation, descriptive statistics, graphical analysis, independent samples t-tests, and one-way ANOVA procedures were performed

Univariate Analysis

Descriptive statistics were calculated for all numerical variables included in the dataset. The table below presents the mean, standard deviation, median, minimum value and maximum value for each variable.

Variable	Mean	SD	Median	Min	Max
cnt	4412.47	1961.11	4400.00	22.00	8714.00
temp1	14.78	8.66	14.05	-3.46	31.87
atemp1	14.69	10.88	14.42	-9.48	39.50
hum1	62.96	14.15	63.08	0.00	97.04
windspeed1	12.83	5.15	12.33	2.83	34.00

The histogram of total bike rentals suggests that the distribution of bike rentals is approximately symmetric, with most observations concentrated between 3,000 and 6,000 rentals.



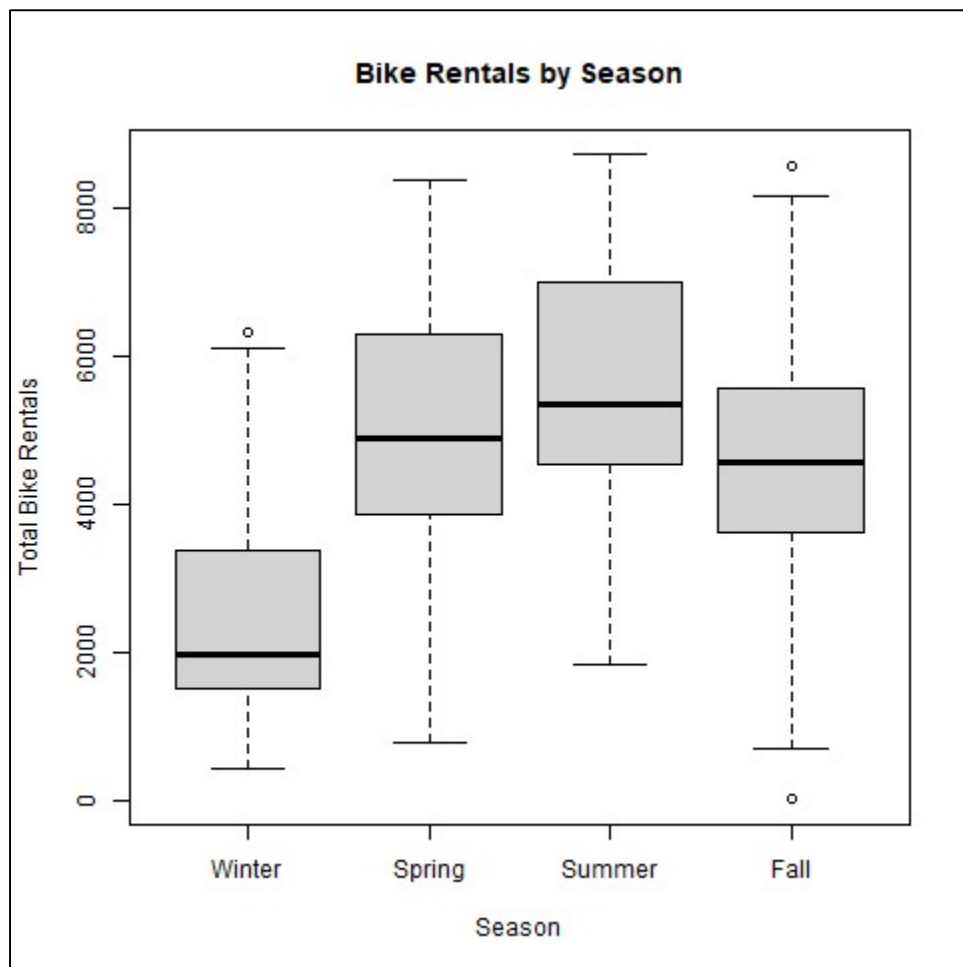
Environmental conditions also varied substantially across observations. Temperatures ranged from winter conditions below freezing to temperatures exceeding 30°C. Humidity and wind speed also showed considerable variation, suggesting that weather-related factors may influence rental demand.

Seasonal Patterns in Bike Rentals

To investigate seasonal differences, average bike rentals were calculated for each season.

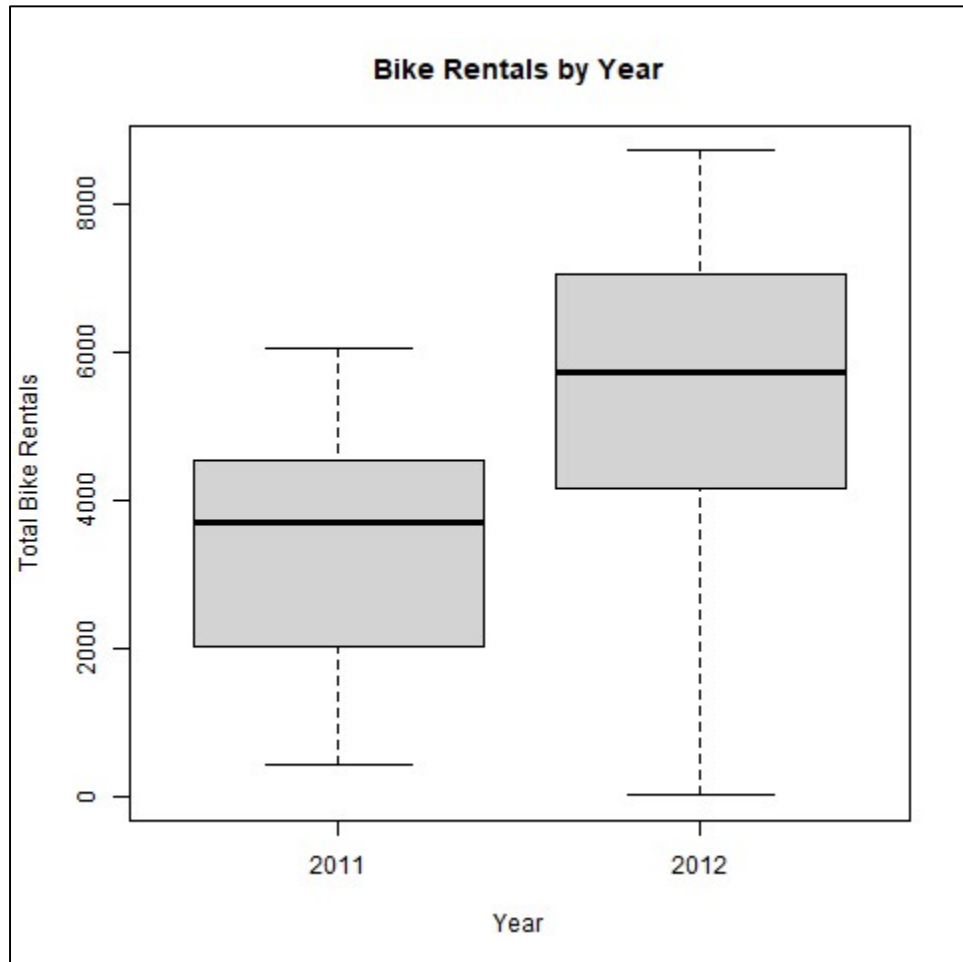
Season	Mean Rentals
Winter	2414.39
Spring	4897.61
Summer	5659.63
Fall	4702.79

The results reveal strong seasonal variation in bike rental demand. Summer recorded the highest average number of rentals, followed by Spring and Fall. Winter recorded substantially lower rental activity than all other seasons.



Comparison of Bike Rentals Between Years

A boxplot was produced to compare bike rentals between 2011 and 2012.



Average rentals increased from 3,344.02 in 2011 to 5,518.93 in 2012. The boxplot indicates that rental activity was consistently higher throughout 2012.

To determine whether this difference was statistically significant, an independent samples t-test was conducted.

Hypothesis

H_0 : The mean number of bike rentals is the same in 2011 and 2012.

H_1 : The mean number of bike rentals differs between 2011 and 2012.

Results

The Welch two-sample t-test produced:

- $t = -15.035$
- $p\text{-value} < 0.001$

Because the p-value is below the 0.05 significance level, the null hypothesis is rejected.

There is a statistically significant difference in bike rental demand between 2011 and 2012. Average rentals were considerably higher during 2012, suggesting substantial growth in the use of the bike-sharing system over time.

Seasonal Differences in Bike Rentals

A one-way ANOVA was conducted to determine whether average bike rentals differed across seasons.

Hypothesis

H_0 : Mean bike rentals are equal across all seasons.

H_1 : At least one season has a different mean number of rentals.

Results

The ANOVA produced:

- $F = 106.4$
- $p\text{-value} < 0.001$

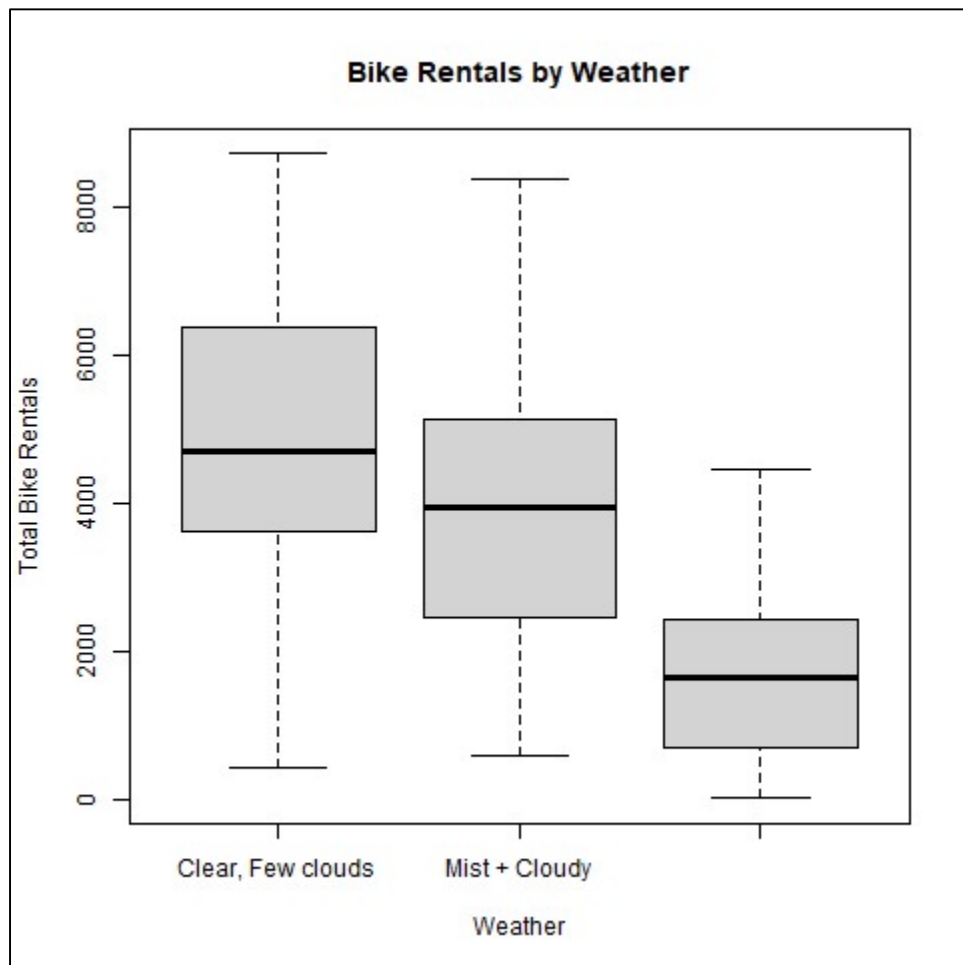
Since the p-value is below 0.05, the null hypothesis is rejected.

The analysis demonstrates statistically significant differences in bike rentals across seasons. Summer recorded the highest rental activity, while Winter recorded the lowest. These findings indicate that seasonal conditions have a substantial effect on bike-sharing demand.

The Tukey post-hoc comparisons further showed that Winter differed significantly from Spring, Summer, and Fall. Summer also differed significantly from both Spring and Fall. The only comparison that was not statistically significant was between Spring and Fall.

Weather Conditions and Bike Rentals

Weather conditions were also examined as a potential determinant of bike rental demand.



The boxplot shows that rental activity decreases as weather conditions worsen. Clear weather conditions are associated with the highest rental volumes, while rainy or snowy conditions are associated with the lowest rental volumes.

A one-way ANOVA was performed to evaluate these differences statistically.

Hypothesis

H_0 : Mean bike rentals are equal across all weather conditions.

H_1 : At least one weather category has a different mean number of rentals.

Results

The ANOVA produced:

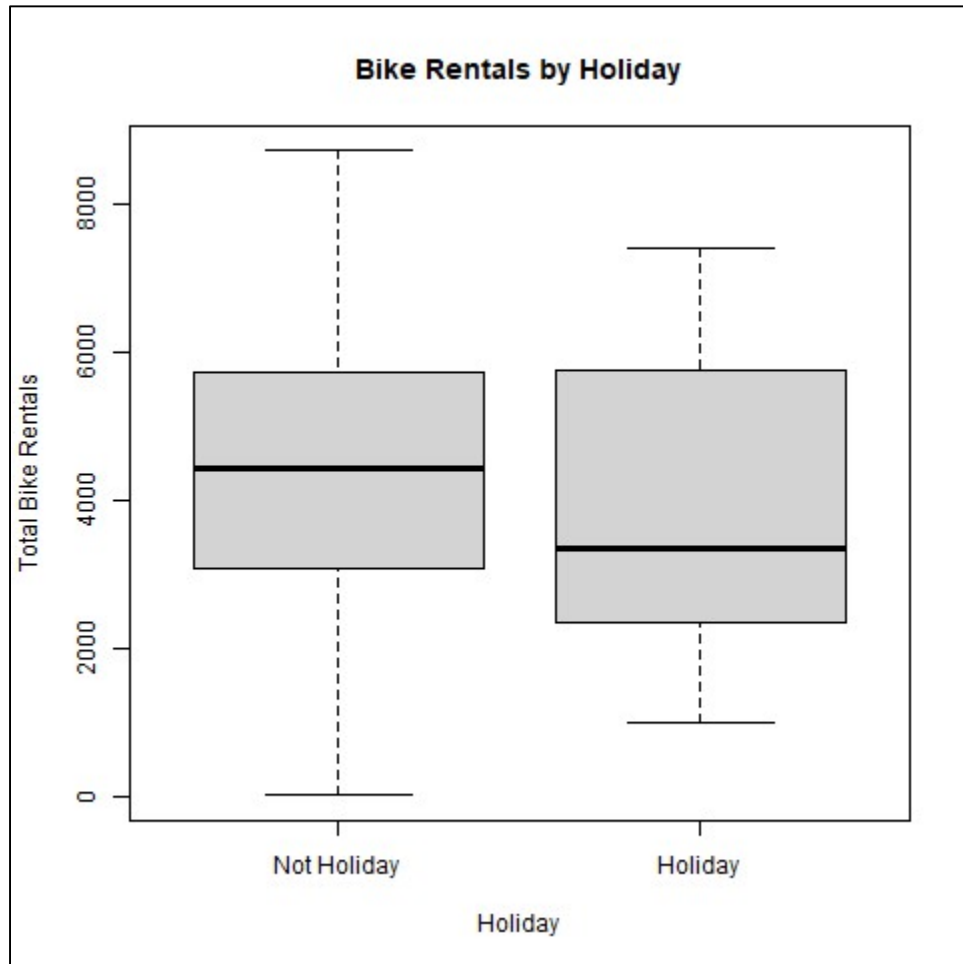
- $F = 29.1$
- $p\text{-value} = 1.07 \times 10^{-12}$

Because the p-value is below 0.05, the null hypothesis is rejected.

Weather conditions have a statistically significant effect on bike rental demand. Better weather conditions are associated with higher rental activity, while poor weather conditions significantly reduce usage.

Holiday Effects

A boxplot comparing holidays and non-holidays was produced.



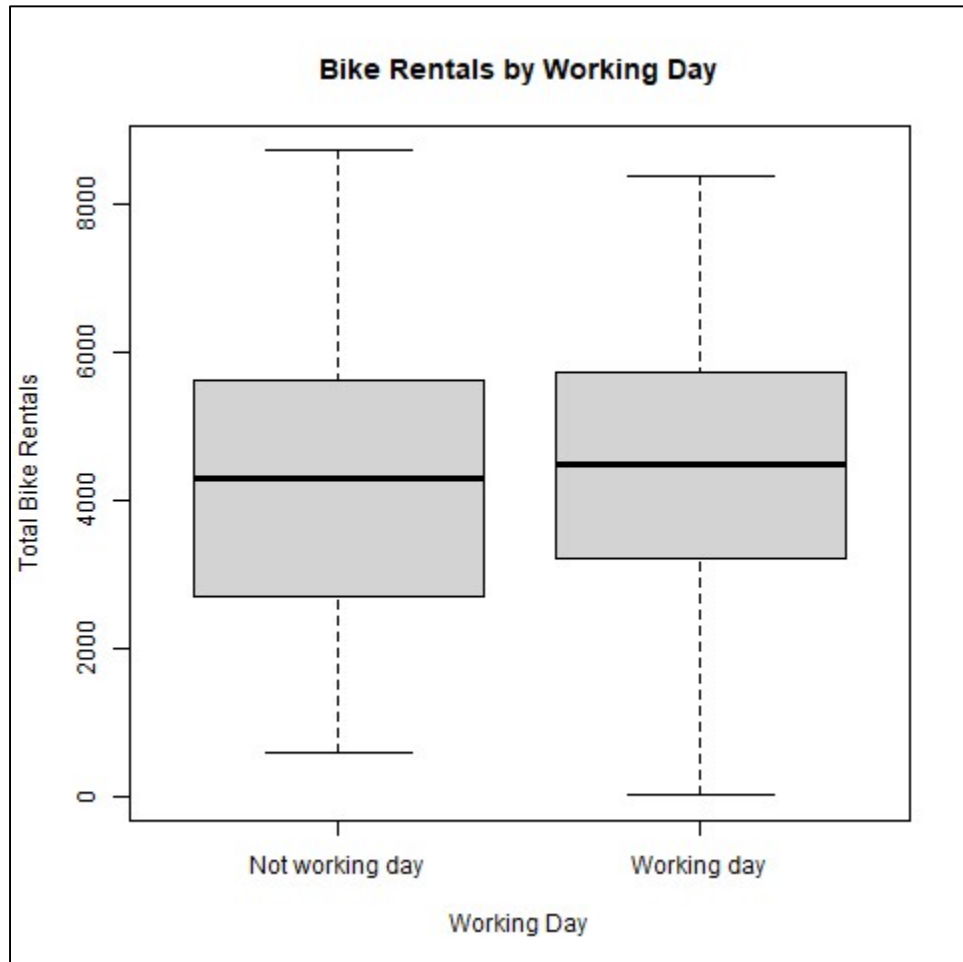
Although average rentals appeared slightly lower on holidays, the difference was relatively small.

An independent samples t-test found no statistically significant difference between holiday and non-holiday rental levels ($p > 0.05$).

Holiday status does not appear to have a significant influence on bike rental demand within this dataset.

Working Day Effects

A boxplot was also produced to compare working days and non-working days.



Rental activity appeared similar across both groups.

An independent samples t-test found no statistically significant difference between working days and non-working days ($p > 0.05$).

Interpretation

The results suggest that working-day status does not significantly affect overall bike rental demand.

Summary of Findings

Several important patterns emerged from the analysis.

First, average bike rental demand increased substantially between 2011 and 2012. This suggests growth in the popularity and adoption of the bike-sharing system over time.

Second, season was identified as one of the strongest determinants of rental demand. Summer recorded the highest average rental volumes, while Winter recorded the lowest.

Third, weather conditions significantly affected bike-sharing activity. Clear weather encouraged higher rental levels, while poor weather conditions reduced usage.

Finally, neither holiday status nor working-day status produced statistically significant differences in rental demand.

Conclusion

This analysis investigated factors associated with bike-sharing demand using a dataset containing 515 observations collected during 2011 and 2012.

The results indicate that bike rental demand is strongly influenced by environmental and seasonal conditions. Significant differences were observed across years, seasons, and weather situations. Rental demand increased considerably during 2012, suggesting growth in system usage. Seasonal effects were substantial, with Summer generating the highest rental activity and Winter generating the lowest. Weather conditions also played an important role, as bike rentals decreased under less favourable weather conditions.

In contrast, holiday status and working-day status did not significantly influence rental demand within the sample analysed.

Overall, the findings suggest that seasonality and weather are the primary drivers of bike-sharing demand and should be considered when planning operational capacity and resource allocation within bike-sharing systems.

References

UCI Machine Learning Repository. Bike Sharing Dataset.

R Core Team (2025). R: A Language and Environment for Statistical Computing.

Wickham, H. ggplot2: Elegant Graphics for Data Analysis.